

Sourav Pal

Member of Technical Staff, Adobe Inc.
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Education

Indian Institute of Technology Kharagpur, India
B.Tech. (Hons.) in Computer Science and Engineering
CGPA: 9.33/10.0

Jul 2014 - May 2018

Research Interests

I am interested in solving challenging problems related to **Machine Learning** and **Computer Vision**. I have worked on applications of **Deep Learning** in Computer Vision and Natural Language Processing.

Papers

- Gupta Prakhar, Shubh Gupta*, Ajaykrishnan Jayagopal*, **Sourav Pal***, and Ritwik Sinha. "Saliency Prediction for Mobile User Interfaces." In Applications of Computer Vision (WACV), 2018 IEEE Winter Conference on, pp. 1529-1538. IEEE, 2018. (Published, [Paper Link](#))
- **Sourav Pal***, Tharun Mohandoss*, Pabitra Mitra. "Visual Attention for Behavioral Cloning in Autonomous Driving." 11th International Conference on Machine Vision (ICMV 2018). (Published, [Paper Link](#))
- Ankan Mullick, **Sourav Pal**, Projjal Chanda, Arijit Panigrahy, Anurag Bharadwaj and Siddhant Singh. "D-FJ: Deep Neural Network Based Factuality Judgment" (Accepted at Truth Discovery and Fact Checking: Theory and practice workshop, ACM SIGKDD 2019)

* Authors contributed equally.

Patents

- Ritwik Sinha, Prakhar Gupta, Shubh Gupta, **Sourav Pal** and Ajaykrishnan Jayagopal. "Saliency Prediction for a Mobile User Interface." US 0159-073001 (Filed in US)
- Prakhar Gupta, Ritwik Sinha, Shubh Gupta, **Sourav Pal** and Ajaykrishnan Jayagopal. "Saliency Prediction for Informational Documents." US 0159-009001 (Filed in US)

Work Experience

Adobe Inc.

Jul'18 - present

Member of Technical Staff, Document Cloud - Adobe Acrobat

- Worked on prototyping a new feature for the Adobe Acrobat Reader. This feature enabled providing intelligent insights around conversations. To this end we used Machine Learning techniques, along with some logical heuristics.
The prototype was declared as the winner of the Adobe Document Cloud HackWeek in Noida.
- Responsible for migration of the build system of the Adobe Acrobat Reader android application from the ndk-build system, to the CMake build system.
- Worked on voice first documents to provide intelligence to the Acrobat Reader and integrating the application with Amazon Alexa.

Internships

Adobe Research, BigData Experience Lab

May'17 - Jul'17

Research Intern

Advisor: Dr. Ritwik Sinha

- Worked on analysis and improvement of mobile user interfaces using eye gaze tracking.
- Used computer vision techniques to develop deep learning models to predict visual saliency of mobile interfaces.
- Developed metrics for evaluating saliency on a new kind of data set, which was collected by deploying an application in Amazon Mechanical Turk.

Presented the work as a talk at the Adobe Tech Summit 2019 (Feb'3-7), San Francisco.

National Digital Library of India, MHRD Government of India.

May'16 - Jul'16

Summer Intern

Advisor: Prof. Partha Pratim Das

- Used dynamic instrumentation for debugging and analysis of multithreaded applications.
- Designed and implemented several algorithms to create tools for data-race detection, deadlock detection, thread-model detection.
- Integrated the above tools with the popular IDE Eclipse as Plug-ins and with Code-blocks as user-defined tools.

Academic Projects

Bachelor's Thesis, Indian Institute of Technology Kharagpur

Jul'17 - May'18

Advisor: Prof. Pabitra Mitra

The goal of the work was to use **Visual Attention** to enhance performance of **Autonomous Vehicles**. The first method proposed a supervised learning approach in which we collected eye-gaze data for the task of driving and used this to train a model for predicting the attention map. The second method described an unsupervised approach where we trained a model to learn to predict attention as it learns to drive a car. The results were conclusive of the fact that incorporation of visual saliency in the autonomous driving agent is capable of improving performance.

The project was selected by the Department of Computer Science & Engineering, IIT Kharagpur for Patent Filing

Artistic Rendering of Images, Machine Learning Term Project

Aug'16 - Nov'16

Advisor: Prof. Pabitra Mitra

Worked on separation of style and content of images and subsequent application of a set of styles to an input image. The solution was first implemented using bi-linear models and then with Convolutional Neural Networks. The CNN model was further extended for doodles and videos.

Detection and Classification of Multiple Objects in an Image

Dec 2017

SMARTathon 2017

Developed an end to end system, which given an image could detect multiple objects and classify them. Proposed novel techniques to improve performance of existing systems. These included an image inpainting approach to better classify overlapping objects, a recursive algorithm which enabled detection and classification of minute objects in an image. Finally confidence scores and correlation between image labels were used to output the most suitable object classes.

Object Segmentation in Images, Image Processing Term Project

Sep'17 - Oct'17

Advisor: Prof. Partha Pratim Das

Worked on detecting the salient boundaries in The Berkeley Segmentation Dataset. Developed an approach based on histogram statistics after initial pre-processing of the original three channel input. The results were comparable to the human annotations.

Deep Learning in opinion detection, Natural Language Processing Term Project

Aug'17-Nov'17

Advisor: Prof. Niloy Ganguly and Mr. Ankan Mullick

Used deep learning techniques to develop models to detect and categorize diverse opinionated sentences from within online news articles, both in general and event specific datasets. Implemented LSTM and CNN based models; used GloVe embedding word vectors and dependency based word embedding for training on MPQA and Yahoo datasets. Also, studied how the sentiment varies according to the opinion fraction in an article.

JuiceLoc, Smart-phone Computing and Applications Term Project

Aug'17-Nov'17

Advisor: Prof. Bivas Mitra

Worked on crowd sourced mobile phone charging point locator in busy locations based on magnetic field strength and WiFi signal strength variations. Developed a neural network model that generates indoor location of available charging stations based on charging patterns of other users.

TinyC Compiler, Compilers Term Project

Jul'16-Nov'16

Advisor: Prof. Partha Pratim Das and Prof. Pralay Mitra

Developed a compiler for a subset of C language. Lexer and parser were built using flex and bison utilities.

Transport Company Computerization Software, Software Engineering Term Project

Mar'16-Apr'16

Advisor: Prof. Partha Pratim Das

Developed a system which can automatically handle the logistics of a transportation company. The system was developed using C++ and MySQL.

Other academic projects, are present in my [Github account](#).

Academic Achievements

- Secured an **All India Rank(AIR) of 411 in JEE Main 2014** out of 1.5 million candidates.
- Secured a rank of **51 in WBJEEM 2014** out of 1,00,000 candidates.

Technical Skills

Programming Languages:

- **Proficient:** Python, C, C++

- **Intermediate:** HTML, Bash, Verilog(HDL), L^AT_EX

Tools: PyTorch, Keras, TensorFlow, Caffe, Numpy, Scikit-learn, OpenCV, Weka, NLTK, Git, CMake

Relevant Coursework

Machine Learning	Deep Learning	Probability and Statistics
Advanced Machine Learning	Matrix Algebra	Image Processing
Artificial Intelligence	Algorithms I and II	Operating Systems
Speech and Natural Language Processing	Software Engineering	Computer Networks
Smartphone Computing and Applications	Database Management Systems	Switching Circuits

Positions of Responsibility

- Served as **School Captain** of The Assembly of God Church School, Purulia. Successfully conducted all student activities during the academic year.
- Served as the **Mentor** of a group of six first year undergraduate students for a period of two years from 2016-2018 at IIT Kharagpur.

Extra Curricular Activities

- Active member of **National Service Scheme(NSS)**. Awarded **Certificate of Recognition** for two years of extraordinary social service.
- One of the 10 teams in India to participate in **SMARTathon India 2017** organized jointly by **Samsung** and **FIIT IIT Delhi**.
- Participated in Inter-hall software development competition, **OpenSoft**. Developed a graph digitizing software using OpenCV
- Participated in **Microsoft** organized hackathon **code.fun.do**, and successfully developed an Android application.
- Emerged as **Runners-up at State Level Inter-school Quiz Competition** held as part of Srijoni 2008.
- Represented Purulia district in Science, Information and Communication Technology Camp 2008-09, organized by **National Council of Science Museums**.